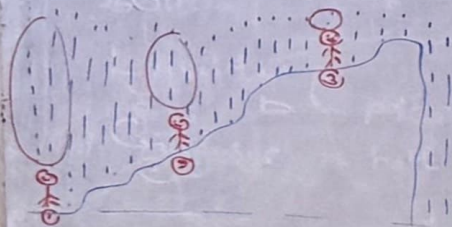


3) Pressure due to atmosphere:

This pressure is caused by air weight (molecules)



The pressure decreases as the altitude increases

Atmospheric Pressure is a function of altitude

$f(x) = x^2$

$$P_1 > P_2 > P_3$$

$$P_3 < P_1 > P_2$$

$$P_{atm} = 10^5 \text{ Pa or } P_{atm} = 1 \text{ atm}$$

ex. 1

A room of $3.5 \text{ m} \times 4.2 \text{ m} \times 2.4 \text{ m}$

a) Find the Air's weight $\rho_{air} = 1.2 \text{ kg/m}^3$

$$W = \rho V \times g = 1.2 \times 35.28 \times 9.8 = 418.4 \text{ N}$$

$$V = 3.5 \times 4.2 \times 2.4 = 35.28 \text{ m}^3$$

b) Find the force of air on a head of $A = 0.04 \text{ m}^2$

$$\because P = \frac{F}{A} \rightarrow F = P \cdot A = 10^5 \times 0.04 = 4000 \text{ N}$$

c) Find the force of air on the floor.

$$F = P \cdot A = 10^5 \times 14.7 = 1470000 \text{ N}$$

$$A = 3.5 \times 4.2 = 14.7 \text{ m}^2$$

3) Pressure due to fluids:

Solids

Solids → Material can not take a shape of a container → Material can keep its shape regardless its container

Liquids

Fluids → Materials can take a shape of a container

Gases

Fluids



$$P_1 = P_2$$

$$P_2 < P_3 > P_1$$

Fluid pressure increases as the depth increases

Fluid is a function of depth

Fluid is at rest → called hydrostatic pressure
 stationary
 not moving

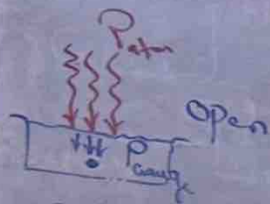
hydraulic

3) Pressure due to fluids:

$$P_{\text{gauge}} = \rho g h$$

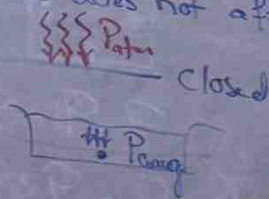
Pressure is direct in all direction (equally)

Pressure does not affected by the shape of the container.



$$P_{\text{abs}} = \rho g h + P_{\text{atm}}$$

Absolute



$$P_{\text{abs}} = \rho g h$$

$$P_{\text{gauge}} = P_{\text{abs}} - P_{\text{atm}}$$

$P_{\text{abs}} > P_{\text{atm}} \rightarrow$ Positive P_{gauge}

$P_{\text{abs}} < P_{\text{atm}} \rightarrow$ Negative $P_{\text{gauge}} \rightarrow$ Suck Pressure

$P_{\text{abs}} = P_{\text{atm}} \rightarrow P_{\text{gauge}} = 0$ (No gauge)